

UBON RATCHATHANI, Thailand

WMO: 484070

Lat: 15.251N

Lon: 104.870E

Elev: 406

StdP: 14.48

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	59.5	62.2	50.3	54.9	68.1	53.2	61.1	69.1	21.0	75.9	19.1	75.4	6.2	20	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	18.7	99.3	78.4	97.5	78.0	95.9	77.7	81.6	90.9	80.9	89.9	80.3	89.1	5.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	154.0	83.9	79.0	152.8	83.8	78.6	150.6	83.6	45.7	91.6	45.1	90.5	44.5	89.6	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.8	13.4	11.5	DB	55.4	102.3	2.6	1.9	53.5	103.6	52.0	104.7	50.5	105.8	48.7	107.1	
			WB	52.5	83.6	2.7	1.5	50.5	84.6	48.9	85.5	47.4	86.3	45.4	87.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.0	76.8	80.3	84.3	86.6	85.4	84.4	83.1	82.8	82.2	81.7	79.7	76.6
	DBStd	4.54	4.10	4.79	4.29	3.55	2.92	2.41	2.38	2.20	2.24	2.26	3.44	4.24
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11676	830	847	1063	1099	1099	1031	1027	1016	967	982	890	825
	CDD65	6201	365	427	598	649	634	581	562	551	517	517	440	360
	CDH74	68802	3558	4801	7502	8510	7694	6824	6179	5944	5308	5172	4116	3194
CDH80	30896	1541	2379	4030	4694	3747	3009	2435	2203	1857	2001	1713	1287	
Wind	WSAvg	4.4	4.8	4.3	3.9	3.7	3.6	4.1	4.4	4.4	3.5	4.3	5.7	6.4
Precipitation	PrecAvg	64.7	0.0	0.4	1.7	3.1	9.3	10.0	12.0	13.0	10.7	3.4	1.1	0.1
	PrecMax	88.1	0.2	2.3	7.1	7.2	25.4	18.9	18.3	22.5	19.6	10.6	3.2	0.7
	PrecMin	50.9	0.0	0.0	0.0	0.4	2.8	5.2	4.7	5.0	6.7	0.2	0.0	0.0
	PrecStd	10.3	0.1	0.7	2.2	1.9	5.7	3.3	4.4	4.2	3.6	2.6	1.0	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.3	97.9	100.7	102.6	100.4	96.9	95.1	93.4	93.1	93.0	94.6	93.5
		MCWB	74.3	76.1	77.8	79.1	79.1	79.3	78.9	80.0	80.0	78.0	77.5	75.7
	2%	DB	92.3	95.6	98.6	100.2	97.5	94.6	92.8	91.6	91.3	91.5	92.6	91.4
		MCWB	73.7	75.4	77.3	78.9	79.3	79.4	79.2	79.5	79.2	77.6	76.6	74.6
	5%	DB	90.3	93.9	96.8	98.2	95.3	92.9	91.1	90.0	89.7	90.0	90.8	89.5
		MCWB	72.7	75.0	76.8	78.8	79.1	79.4	78.8	79.0	78.8	77.2	76.0	73.7
	10%	DB	88.2	91.7	94.9	95.8	93.2	91.2	89.5	88.6	88.1	88.9	88.8	87.3
		MCWB	71.7	74.3	76.5	78.4	78.9	79.1	78.5	78.8	78.6	77.0	75.3	72.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.9	79.6	81.0	82.5	81.9	81.9	81.5	81.8	81.7	80.7	80.0	77.9
		MCDB	89.2	91.5	94.6	96.3	93.1	90.2	88.6	88.4	88.0	88.0	88.9	89.4
	2%	WB	76.2	78.3	79.5	81.0	81.0	81.0	80.5	80.8	80.6	79.7	78.9	76.5
		MCDB	87.5	91.0	93.3	94.5	91.6	89.3	88.0	87.5	87.0	86.9	88.0	87.7
	5%	WB	74.8	77.2	78.6	80.1	80.3	80.3	79.8	80.0	79.9	79.0	78.0	75.3
		MCDB	86.2	89.8	92.4	93.2	90.4	88.6	87.1	86.6	86.1	86.2	86.7	86.2
	10%	WB	73.5	76.1	77.8	79.3	79.7	79.8	79.2	79.5	79.2	78.3	76.9	74.0
		MCDB	85.2	87.8	90.9	91.7	89.2	87.7	86.3	85.8	85.2	85.3	85.2	84.7
Mean Daily Temperature Range	5% DB	MDBR	23.1	23.0	21.2	18.7	15.8	13.8	12.5	11.9	12.2	14.9	18.0	20.6
		MCDBR	24.2	23.3	22.3	20.7	18.5	16.0	14.8	13.8	14.5	16.7	19.7	21.9
		MCWBR	8.5	7.6	7.0	5.7	5.2	5.0	4.8	4.9	5.0	5.9	6.8	7.8
	5% WB	MCDBR	22.2	20.5	20.3	19.2	16.4	14.7	13.5	12.8	13.2	14.8	17.8	20.5
MCWBR		8.4	7.7	7.1	6.1	5.5	5.3	5.2	5.3	5.2	5.7	6.6	8.0	
Clear-Sky Solar Irradiance	taub	0.463	0.587	0.665	0.611	0.500	0.466	0.462	0.462	0.494	0.505	0.452	0.443	
	taud	2.150	1.818	1.685	1.851	2.189	2.299	2.309	2.302	2.201	2.145	2.277	2.247	
	Ebn at Noon	262	234	220	233	256	263	264	267	259	253	264	265	
	Edh at Noon	47	68	80	68	47	42	42	43	47	48	41	42	
All-Sky Solar Radiation	RadAvg	1633	1720	1690	1770	1744	1617	1520	1459	1456	1596	1586	1554	
	RadStd	80	78	56	80	87	86	131	101	135	115	65	57	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	+0.52	N/A	N/A	N/A	+0.40	+0.36	N/A	N/A	+185	+184

Nomenclature: See separate page